Retail Sales Patterns and Prediction Using FreshRetailNet 50K

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1 Dataset and Source

Our group will use FreshRetailNet-50K, published by Dingdong-Inc on Hugging Face. The dataset documentation describes 4.5 million training records and 350,000 evaluation records. Each daily record identifies a store, product, and date and contains scaled sales, promotional discounts, holiday and marketing indicators, stockout information, and weather measurements. Hourly sales and stock-status information are also provided.

To keep the project manageable, we propose selecting two cities, five stores per city, and ten product series per store. Retaining 90 training days for 100 complete store-product combinations would produce approximately 9,000 training records. The exact combinations and final record counts will be confirmed after checking coverage and data quality. We will document the selection procedure and retain complete histories rather than sampling isolated daily records.

The sales values are normalized and will be interpreted as relative sales, not actual revenue or unit counts. Observed sales during stockouts may understate customer demand. Encoded location and product identifiers also limit business interpretation.

2 Research Question

Which promotional, calendar, weather, and stockout factors are associated with observed retail sales, and can information available in advance improve next-day sales predictions beyond using past sales alone?

This question connects the team's interest in retail demand with practical inventory and promotion planning. Exploratory analysis will describe sales relationships, while predictive modeling will evaluate whether selected information improves forecasts. Findings will be interpreted as associations and predictive contributions rather than proof that a factor causes a change in sales.

3 Proposed Methodology

Data preparation. We will inspect the selected data for missing values, duplicate store-product-date records, inconsistent types, and gaps in daily coverage. Dates will be converted, identifiers treated as categorical variables, and past-sales and calendar features created as appropriate. We will preserve zero-sales and stockout observations unless a documented data-quality issue warrants a correction. The preparation steps will be recorded even if little cleaning is needed.

Exploratory analysis. We will examine sales distributions and trends, compare promotional and nonpromotional periods, and summarize sales alongside holidays, weather, and stockout hours. Comparisons across selected stores and products will help identify useful variation and refine the predictors. Analysis will distinguish recorded sales from demand that could not be fulfilled.

Predictive methods. The proposed approach will compare a previous-week same-day sales baseline with linear regression and a Random Forest regression model. The baseline provides a simple reference, regression supports interpretation, and Random Forest allows nonlinear relationships. Final model settings and features will be selected using training and validation data. Python is planned for preparation and modeling, with visualization tools selected according to team needs.

Evaluation. Training, validation, and testing will follow chronological order. We will verify the published split dates before assigning evaluation periods and preserve a separate final test set. Preprocessing will be fitted using training data only. Next-day predictions will use only information available at forecast time; realized target-day sales, stockouts, and weather will not be treated as advance predictors. Historical weather and stockout information may be used, and future promotion or holiday indicators only when known in advance. We propose mean absolute error and root mean squared error to compare performance on the same held-out observations.

Interpretation and limitations. The report will discuss forecast performance, informative factors, and implications for inventory and promotion decisions. A 90-day history cannot establish annual seasonality. The restricted sample and anonymized identifiers limit generalization. The project will focus on observed sales rather than claiming to measure the full amount of unmet demand.

Reference

Dingdong-Inc. (2025). FreshRetailNet-50K [Data set]. Hugging Face. https://huggingface.co/datasets/Dingdong-Inc/FreshRetailNet-50K